Dobinski's Formula for Bell Numbers

Alethfeld Proof System

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Theorem 0.1 (Dobinski's Formula). *For all non-negative integers n, the Bell number B_n equals*

$$B_n = \frac{1}{e} \sum_{k=0}^{\infty} \frac{k^n}{k!}$$

where B_n counts the number of partitions of a set with n elements.

Proof of Theorem **??**. **1.** By definition, $B_n = \sum_{k=0}^{n} S(n,k)$, the sum of Stirling numbers of the second kind. ✓

1.1. For $n \geq 1$, every partition of an n-set has exactly k non-empty blocks for some unique k with $1 \leq k \leq n$. For $n = 0$, the unique partition (the empty partition) has $k = 0$ blocks. ✓

1.2. The set of all partitions of an n-set is the disjoint union of partitions with exactly k blocks, for $k = 1, 2, \ldots, n$. ✓

1.3. By definition of $S(n,k)$, there are exactly $S(n,k)$ partitions with exactly k blocks. ✓

1.4. By the addition principle for disjoint sets, $B_n = \sum_{k=1}^{n} S(n,k)$. Since $S(n,0) = 0$ for $n > 0$, we can write $B_n = \sum_{k=0}^{n} S(n,k)$. ✓

2. $S(n,k)$ counts surjections from an n-set to a k-set divided by $k!$. Equivalently:

$$S(n,k) = \frac{1}{k!} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} j^n$$

✓

3. Substituting the explicit formula:

$$B_n = \sum_{k=0}^{n} \frac{1}{k!} \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} j^n$$

✓

4. Exchanging the order of summation and extending k to infinity (terms with $k > n$ contribute 0 to inner sum when $j \leq n$):

$$B_n = \sum_{j=0}^{\infty} j^n \sum_{k=j}^{\infty} \frac{(-1)^{k-j}\binom{k}{j}}{k!}$$

✓

4.1. For the finite double sum $\sum_{k=0}^{n}\sum_{j=0}^{k} f(j,k)$, we can exchange order of summation to $\sum_{j=0}^{n}\sum_{k=j}^{n} f(j,k)$ since both are finite sums over the same triangular region $\{(j,k) : 0 \le j \le k \le n\}$. ✓

4.2. After exchange: $B_n = \sum_{j=0}^{n} j^n \sum_{k=j}^{n} \frac{(-1)^{k-j}\binom{k}{j}}{k!}$ ✓

4.3. For fixed j, the inner sum $\sum_{k=j}^{\infty} \frac{(-1)^{k-j}\binom{k}{j}}{k!}$ converges absolutely (by ratio test), so we can extend the finite sum to infinity. ✓

4.4. Therefore $B_n = \sum_{j=0}^{\infty} j^n \sum_{k=j}^{\infty} \frac{(-1)^{k-j}\binom{k}{j}}{k!}$ with the extension being well-defined. ✓

4.1. Powers can be written in terms of Stirling numbers: $k^n = \sum_{j=0}^{n} S(n,j)\cdot(k)_j$ where $(k)_j = k(k-1)\cdots(k-j+1)$ is the falling factorial. Equivalently, $k^n = \sum_{j=0}^{n} S(n,j)\cdot j!\cdot\binom{k}{j}$ where $\binom{k}{j} = 0$ for $k < j$. ✓

4.2. Substituting into Dobinski's sum:

$$\frac{1}{e}\sum_{k=0}^{\infty}\frac{k^n}{k!} = \frac{1}{e}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{j=0}^{n} S(n,j)\cdot j!\cdot\binom{k}{j} = \frac{1}{e}\sum_{j=0}^{n} S(n,j)\cdot j!\cdot\sum_{k=j}^{\infty}\frac{\binom{k}{j}}{k!}$$

✓

4.3. For $k \ge j$: $\frac{\binom{k}{j}}{k!} = \frac{1}{j!\cdot(k-j)!}$. So $\sum_{k=j}^{\infty}\frac{\binom{k}{j}}{k!} = \frac{1}{j!}\sum_{m=0}^{\infty}\frac{1}{m!} = \frac{e}{j!}$ ✓

4.4. Therefore $\frac{1}{e}\sum_{k=0}^{\infty}\frac{k^n}{k!} = \frac{1}{e}\sum_{j=0}^{n} S(n,j)\cdot j!\cdot\frac{e}{j!} = \sum_{j=0}^{n} S(n,j) = B_n$ ✓

5. The inner sum $\sum_{k=j}^{\infty}\frac{(-1)^{k-j}\binom{k}{j}}{k!} = \frac{1}{e\cdot j!}$ by the series expansion of e^{-1}. ✓

6. Substituting:

$$B_n = \sum_{j=0}^{\infty} j^n \cdot \frac{1}{e\cdot j!} = \frac{1}{e}\sum_{j=0}^{\infty}\frac{j^n}{j!}$$

✓

7. The series $\sum_{k=0}^{\infty}\frac{k^n}{k!}$ converges absolutely for all $n \ge 0$ by ratio test comparison with e^k. ✓

8. Therefore $B_n = \frac{1}{e}\sum_{k=0}^{\infty}\frac{k^n}{k!}$, which is Dobinski's Formula. □ ✓

□